

	UNIVERSITY OF ADEN - FACULTY OF ENGINEERING			
	Odd SEMESTER EXAMINATION, ACAD.YEAR 2024-2025			
	Student Name			Reg. No
	Subject	Principle of communication systems		
	Class	B4E /E	Date	20-01-25
	Max. Marks	50	Time	2 hours
Lecturer	Eng .Shada A.Adeem		Signature	

- NOTES:** 1) QUESTION ONE IS COMPULSORY.
 2) ANSWER ANY other **THREE** QUESTIONS.
 3) They can use calculator

QUESTION №1 (Compulsory)

1. Square Law modulators are	
A. used for frequency modulation	B. used for pulse width modulation
used for amplitude modulation	D. used for phase modulation
2. If the synchronization between the transmitter and receiver fails which of the following pulse system will be affected	
A. PAM	PWM
C. All of them	PPM
3. Which multiplexing technique used to transmit digital signals?	
A. TDM	B. FDM
A&B	D. None of them
4. What is the reason of "envelope" in an amplitude modulated signal? To determine.....	
A. noise signal	baseband signal
C. carrier signal	D. none of them
5. For getting 100% modulation, carrier amplitude should	
A. exceed signal amplitude	be equal to signal amplitude
be equal to 0	D. be lesser than signal amplitude
6. The minimum channel Bandwidth is used by which modulation technique?	
A. VSB	B. DSB-SC
C. AM	D. DSB-FC
7. In a PCM signal ,if we increase the quantization level from 2 to 8 , how do the relative bandwidth vary	
A. Doubled	Tripled
C. remain the same	D. Eight times
8. What is the maximum transmission efficiency?	
A. 67.88%	33.33%
C. 73%	D. 30.33%
9. What is the total gain of amplifiers with a power gain of 6, 8, 5 when connected in series?	
A. 19	B. 20
C. 190	D. 240
10. Singleton amplitude modulation	
A. consists of only one frequency component	contains a large number of frequency components
C. contains infinite number of frequency components	D. contains no frequency components
11. Ring Modulator is	
A. used for DSB-SC generation	B. used for SSB-SC generation
C. used for VSB generation	D. used for AM generation

12. In _____, an analog signal is sampled at equal intervals, with the resulting pulses still analog in value	
A. PCM	B. PAM
C. FM	D. DSB-SC
13. When aliasing will take place?	
A. Sampling signals less than Nyquist Rate	B. Sampling signals more than Nyquist Rate
C. Sampling signals equal to Nyquist Rate	D. Sampling signals at a rate which is twice of Nyquist Rate
14. Overmodulation results in _____	
A. Distortion	B. Weakens signal
C. Strengthens the signal	D. provides immunity to noise
15. In _____, the first step after PAM is quantization of the analog pulses.	
A. PWM	B. PPM
C. PCM	Demodulator

QUESTION № 2

- What is the sampling theorem In detailed ,show the concept of under sampling & over sampling, with necessary waveforms?
- Draw the block diagram of Basic Elements of PCM ?
- A carrier wave $6\cos(2\pi \times 600 \times 10^3 t)$ volts , is used to modulate an audio signal given by:
 $m(t) = 0.3\cos(2\pi \times 1000t) + 0.2\cos(2\pi \times 2000t) + 0.1\cos(2\pi \times 3000t)$ volts
Determine the:
 - Upper and lower sideband frequencies.
 - Sketch the spectrum of the modulated signal.
 - Find the total power in the sidebands.

QUESTION № 3

- Explain the generation and recovery of the PAM (Flat Top) signal ,with necessary equations and spectrums diagrams ?
- Compare between NB-FM (Narrow band)and WB-FM (Wide band)?
- For the given analog signal : $x(t) = 3\cos(100\pi t) + 10\sin(600\pi t) + \cos(800\pi t)$
Using **FDM(Frequency Division Multiplexing)** system with carrier frequency :
 $f_{c1} = 300\text{Hz}$ - (for the $3\cos(100\pi t)$ signal)
 $f_{c2} = 700\text{Hz}$ - (for the $10\sin(600\pi t)$ signal)
 $f_{c3} = 1500\text{Hz}$ - (for the $\cos(800\pi t)$ signal) ,
Determine the:
 - bandwidth of each modulating signal
 - Total bandwidth and guard band between signals
 - Draw the frequency spectrum .

QUESTION № 4

- Explain the generation of the PWM signal ,with necessary block diagram and waveforms representation of signal generation ?
- Compare between AM and FM ?
- An FM wave is expressed as:

$$x(t) = 30\cos(2\pi \times 10^6 t + 5\sin(2\pi \times 2000t))$$

Determine : The Frequency Deviation , the Bandwidth and the Power of the FM wave ?

QUESTION № 5

- a) What are the main different between AM-DSB.FC , AM-SSB , AM-VSB .In power transmission ,bandwidth ,waveform representation and equation representation (note :put your answer in table if you could) ?
- b) Draw the block diagram of Costas receiver and write short notes how we can use it to demodulate DSB-SC?
- c) An audio signal represented by $15\sin(300\pi t)$ is utilized to amplitude modulate a carrier wave of $80\sin(2000\pi t)$. Assume the modulation index is 0.5 Calculate:
1. The frequencies of the sidebands.
 2. The amplitude of each sideband.
 3. The efficiency of the amplitude modulation.
 4. sketch of the modulated waveform and frequency spectrum

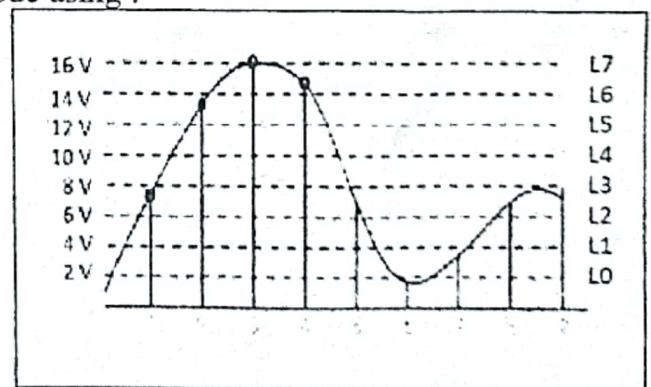
QUESTION № 6

- a) What is multiplexing why we need it, then draw the block diagram of FDM ,TDM ?
- b) Give four reasons why we need digitize analogue source ?
- c) For the given figure complete the table (With uniform quantization):

Sampled values of Analog signal	7.7 v	13.7 v	16 v	14.3v	6.8 v
Nearest quantizing voltage					
Quantization error					
Binary code					

Then : From the binary code draw the line code using :

- 1.Unipolar NRZ
- 2.Polar RZ



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	UNIVERSITY OF ADEN - FACULTY OF ENGINEERING		
	Odd Semester RE-EXAMINATION, ACADEMIC YEAR 2014-2015		
	Student Name	Reg. No.	
	Subject	Principle of communication systems	
	Class		
	Max. Marks		
	Lecturer		

NOTE: 1) ANSWER ANY FOUR QUESTIONS

2) They can use calculator

QUESTION No.1

- Explain the generation of the PPM signal, with necessary block diagram and waveforms representation of signal generation?
- Mention the advantages and disadvantages of the Frequency Modulation?
- An audio signal represented by $m(t) = 10\sin(400\pi t)$ Volt, is utilized to amplitude modulate a carrier wave of: $x_c(t) = 50\sin(2500\pi t)$ Volt, Assuming the modulation index is 0.8, calculate:

- The frequencies of the sidebands.
- The amplitude of each sideband.
- The efficiency of the amplitude modulation.
- A sketch of the modulated waveform and frequency spectrum.

QUESTION No.2

- What is the Quantization in details, why we need it after sampling, explain your answer with your example?
- Draw the generation block diagram of AM (DSB-FC) with short notes about each block?
- An analog signal is expressed by the equation:

$$m(t) = 25\cos(400\pi t) + 15\sin(600\pi t) + \cos(1200\pi t)$$

Calculate:

- The frequencies of the components in the signal.
- The maximum frequency (f_{max}) among the components.
- The Nyquist rate and Nyquist interval for the signal.

QUESTION No.3

- What are the advantages and disadvantages of AM-SSB (single side band)?
- Explain the generation and recovery of the PAM (Flat Top) signal, with necessary equations and spectrum diagrams?
- For the given analog signal: $x(t) = 3\cos(100\pi t) + 10\sin(600\pi t) + \cos(800\pi t)$
Using FDM (Frequency Division Multiplexing) with carrier frequency:

$$f_{c1} = 300 \text{ Hz} \text{ - (for the } 3\cos(100\pi t) \text{ signal)}$$

$$f_{c2} = 700 \text{ Hz} \text{ - (for the } 10\sin(600\pi t) \text{ signal)}$$

$$f_{c3} = 1500 \text{ Hz} \text{ - (for the } \cos(800\pi t) \text{ signal),}$$

Determine the:

- bandwidth of each modulating signal
- Total bandwidth and guard band between signals
- Draw the frequency spectrum.

QUESTION No. 4

- Explain the Detection of the PWM signal ,with necessary block diagram and waveforms representation of signal generation ?
- Compare between AM and FM ?
- An FM wave is expressed as:

$$x(t) = 30 \cos(2\pi \times 10^6 t + 5 \sin(2\pi \times 2000 t))$$

Determine : The Frequency Deviation , the Bandwidth and the Power of the FM wave ?

QUESTION No. 5

- Draw the block diagram of basic elements of PCM?
- What are the main different between AM-DSB,FC , AM-SSB .In power transmission ,bandwidth ,waveform representation and equation representation ?
- An angle-modulated signal is represented by the equation:

$$S(t) = 12 \cos(2\pi \times 1.2 \times 10^6 t + 7 \sin(2\pi \times 3000 t)) \text{ volts}$$

Find the following:

- The power in the modulated signal.
- The frequency deviation.
- The deviation ratio.
- The transmission bandwidth.

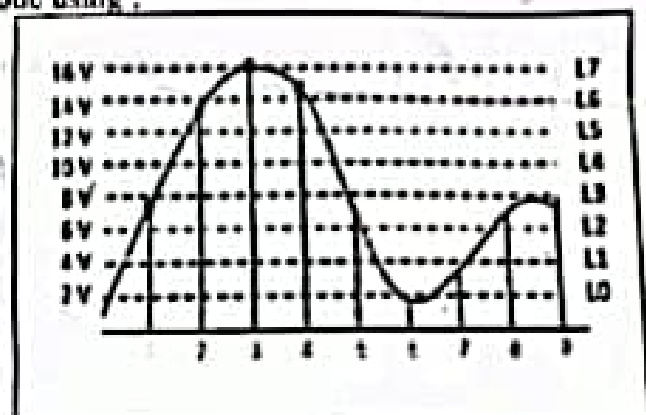
QUESTION No. 6

- What are the advantages of Modulation?
- What is multiplexing why we need it, then draw the block diagram of FDM ,TDM ?
- For the given figure complete the table (With uniform quantization):

Sampled values of Analog signal	7.7 v	13.7 v	16 v	14.3 v	6.8 v
Nearest quantizing voltage					
Quantization error					
Binary code					

Then : From the binary code draw the line code using :

- polar NRZ
- Polar RZ



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		UNIVERSITY OF ADEN - FACULTY OF ENGINEERING	
		Even SEMESTER EXAMINATION, ACAD. YEAR 2024-2025	
Student Name	فتاة سعيدة	Reg. No	2405072179
Subject	Principle of communication		
Class	B2IT/E	Date	12.9.25
Max. Marks	50	Time	2 hours
Lecturer	Eng. Shada A. Adeem		Signature

NOTES: 1) ANSWER ANY FOUR QUESTIONS.

2) Calculators are allowed.

QUESTION №1

- Draw the block diagram- of basic elements of PCM (Transmitter / channel / receiver)?
- Compare between FSK and PSK modulation, in any five points ?
- An audio signal : $20 \sin 2\pi(3000t)$, amplitude modulates a carrier $100 \sin 2\pi(150,000t)$, then :
 - Calculate the modulation index
 - Write the general equation for the modulated wave
 - Calculate The efficiency of the amplitude modulation
 - sketch frequency spectrum.

QUESTION №2

- Draw the wave forms representation of Vestigial Side Band Suppressed Carrier (VSBSC) and write down its modulated equation, the bandwidth and power equations?
- Compare between Synchronous TDM and Asynchronous, in five points?
- Consider an analog signal defined by the equation:

$$x(t) = 10\sin(500\pi t) + 25\sin(900\pi t) + \sin(1800\pi t)$$
 Answer the following questions about this signal:
 - What are the frequencies of each of the sinusoidal components?
 - What is the maximum frequency present in the signal?
 - What is the minimum sampling rate required to avoid aliasing (Nyquist-rate)?
 - What is the corresponding sampling interval?

QUESTION №3

- What are the two types of Companding ,draw the compression graph of one of them and why we need a compounding ?
- Draw the generation block diagram of Coherent Detector (DSB-SC Demodulators)?
- For a given analog signal, $x(t) = 5\sin(40\pi t) + 8\cos(300\pi t) + 2\sin(500\pi t)$, we use Frequency Division Multiplexing (FDM) with the following carrier frequencies:

$$f_{c1} = 500 \text{ Hz (for the } 5\sin(40\pi t) \text{ signal) , } f_{c2} = 1200 \text{ Hz (for the } 8\cos(300\pi t) \text{ signal) , } f_{c3} = 2000 \text{ Hz (for the } 2\sin(500\pi t) \text{ signal)}$$

Determine the following:

- The bandwidth of each modulating signal.
- The total bandwidth of the multiplexed signal and the guard band between each channel.
- Draw the frequency spectrum of the multiplexed signal.

QUESTION № 4

- Compare between Digital modulation and Analog Modulation, in five points?
- Draw how AM-modulated signal will look like when modulation index
($\mu > 1$, $\mu < 1$, $\mu = 1$), which value is the best?
- A frequency-modulated wave has a carrier amplitude of 20 V, a carrier frequency of 12 MHz, and a modulating signal with a frequency of 3 kHz. The modulation index is 7. Calculate the following for this FM wave:
 - Frequency Deviation (Δf)
 - Bandwidth (BW)
 - Power (P)

QUESTION № 5

- Draw data flow between transmitter and receiver of Synchronous TDM and Asynchronous TDM?
- Compare between NB-FM(narrow band) and WB-FM(wide band)?
- An angle-modulated signal is represented by the equation:
$$S(t) = 25 \cos(2\pi \times 2 \times 10^6 t + 5 \sin(2\pi \times 5000 t))$$
 volts.

Find the following:

- The power in the modulated signal.
- The frequency deviation.
- The deviation ratio.
- The transmission bandwidth.

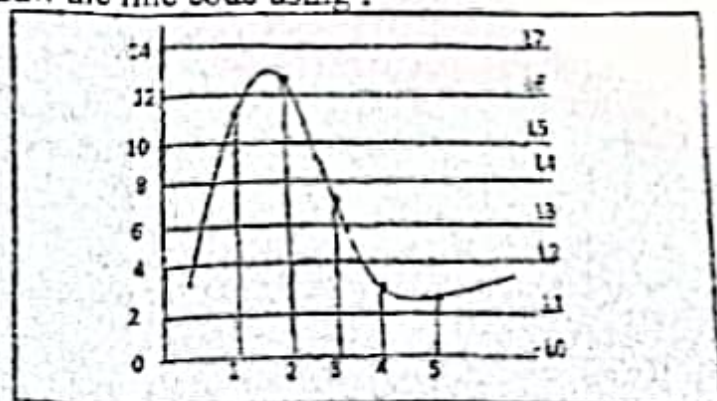
QUESTION № 6

- Draw the block diagram of a modulation and demodulation of the ASK (amplitude shift key) modulation?
- Compare between AM modulation and FM modulation?
- For the given figure complete the table (With uniform quantization):

Sampled values of Analog signal	11.2v	13.1 v	7.2v	3.1v	2.3 v
Nearest quantizing voltage					
Quantization error					
Binary code					

Then : From the bit sequences draw the line code using :

- Polar NRZ
- Bipolar RZ



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Even SEMESTER RE- EXAMINATION, ACAD. YEAR 2024-2025			
Student Name			Reg. No
Subject	Principle of communication		
Class	B211/E	Date	
Max. Marks	50	Time	2 hours
Lecturer	Eng. Shada A. Adeem	Signature	

NOTES: 1) ANSWER ANY FOUR QUESTIONS.

2) They can use calculator

QUESTION No 1

- Mention the advantages and disadvantages of the Frequency Modulation ?
- An audio signal represented by $m(t) = 10\sin(400\pi t)$ Volt, is utilized to amplitude modulate a carrier wave of: $X_c(t) = 50\sin(2500\pi t)$ Volt, Assuming the modulation index is 0.8, calculate:

- The frequencies of the sidebands.
- The amplitude of each sideband.
- The efficiency of the amplitude modulation.
- A sketch of the modulated waveform and frequency spectrum.

$$M = \frac{A_m}{A_c}$$

QUESTION No 2

- Draw the generation block diagram of AM (DSB-FC) ?
- An analog signal is expressed by the equation:

$$m(t) = 25\cos(400\pi t) + 15\sin(600\pi t) + \cos(1200\pi t)$$

- The frequencies of the components in the signal.
- The maximum frequency (f_{max}) among the components.
- The Nyquist rate and Nyquist interval for the signal.

$$\eta = \frac{m^2}{2 + m^2} \times 100$$

$$f_c =$$

QUESTION No 3

- What are the advantages of AM-SSB (single side band) ?
- An analog signal is expressed as:

$$x(t) = 5\cos(\pi(1500)t) + 8\cos(\pi(6000)t) + 15\cos(\pi(20000)t)$$

This signal is used in a frequency multiplexing system (FDM), with carrier frequencies of 25 kHz, 35 kHz, and 50 kHz, respectively.

- Find the bandwidth of each modulating signal.
- Calculate the total bandwidth required for the multiplexed system.
- Determine the guard band needed to avoid interference between the channels.
- Draw the frequency spectrum of the system, labeling each component clearly.
- Is it a single tone signal or a multitone signal? Justify your answer.

$$f = \frac{B}{2\pi}$$

$$B = 2f_m$$

$$f_m = \frac{\omega_m}{2\pi} = 15$$

$$25 \times 10^3$$

QUESTION № 4

a) Compare between AM and FM ?

b) An FM wave is expressed as:

$$x(t) = 30 \cos(2\pi \times 10^6 t + 5 \sin(2\pi \times 2000 t))$$

Determine : The Frequency Deviation , the Bandwidth and the Power of the FM wave ?

$$\Delta f = \frac{f_m}{f_c}$$

QUESTION № 5

a) Draw the block diagram of basic elements of PCM?

b) An angle-modulated signal is represented by the equation:

$$S(t) = 12 \cos(2\pi \times 1.2 \times 10^6 t + 7 \sin(2\pi \times 3000 t)) \text{ volts}$$

Find the following:

1. The power in the modulated signal.
2. The frequency deviation.
3. The deviation ratio.
4. The transmission bandwidth.

$$P =$$

$$\Delta f_m = \beta f_m$$

$$\beta = \frac{\Delta f_m}{f_m} = 2 \Delta f_m$$

QUESTION № 6

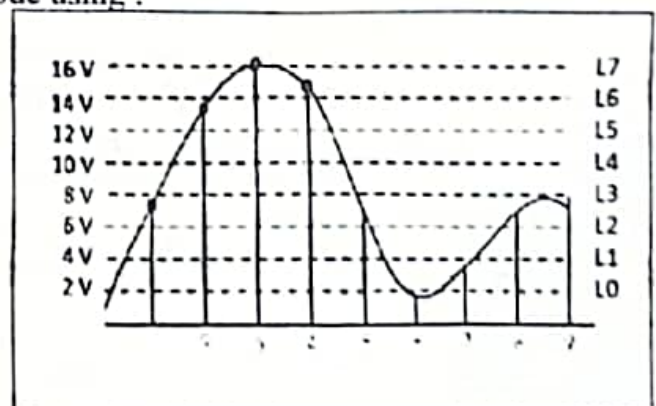
a) Compare between NB-FM (Narrow band)and WB-FM (Wide band)?

b) For the given figure complete the table (With uniform quantization):

Sampled values of Analog signal	7.7 v	13.7 v	16 v	14.3v	6.8 v
Nearest quantizing voltage					
Quantization error					
Binary code					

Then : From the binary code draw the line code using :

1. polar NRZ
2. Polar RZ



NB/FMA

Modulated index
Bandwidth

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Number frequency
Frequency

Application